

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

8MA0/22

Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

Mock Set 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A car is moving along a straight road.

- At time $t = 0$, the speed of the car is 18 m s^{-1}
- The car decelerates at a constant rate for 5 seconds from time $t = 0$, reducing its speed from 18 m s^{-1} to $V \text{ m s}^{-1}$
- The car then continues at a constant speed of $V \text{ m s}^{-1}$ for 15 seconds
- The car then accelerates at a constant rate for 5 seconds, increasing its speed from $V \text{ m s}^{-1}$ to 16 m s^{-1}

(a) Sketch a speed-time graph for the motion of the car from $t = 0$ to $t = 25$ seconds. (2)

Between $t = 0$ to $t = 25$ seconds, the car travels a total distance of 325 m.

(b) Find the value of V . (4)

(c) Find the deceleration of the car. (2)



DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 8 marks)



2. A particle P moves along the positive x -axis.

At time t seconds, $t \geq 0$

- the distance of P from the origin O is x metres
- the velocity of P is $v \text{ ms}^{-1}$ in the positive x direction
- the acceleration of P is $a \text{ ms}^{-2}$

Given that $x = 4t^3 - 21t^2 + 36t + 1$

(a) find v in terms of t , (2)

(b) find the distance of P from O at the instant when P is at rest for the second time, (4)

(c) find a in terms of t . (2)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass m kg is moving on a smooth horizontal plane.

The particle P is moving under the action of two forces $(2\mathbf{i} - \mathbf{j})\text{N}$ and $(c\mathbf{i} - 13.4\mathbf{j})\text{N}$, where c is a constant.

The acceleration of P is $(10\mathbf{i} - 24\mathbf{j})\text{ms}^{-2}$

Find the value of c .

(4)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 4 marks)



4.

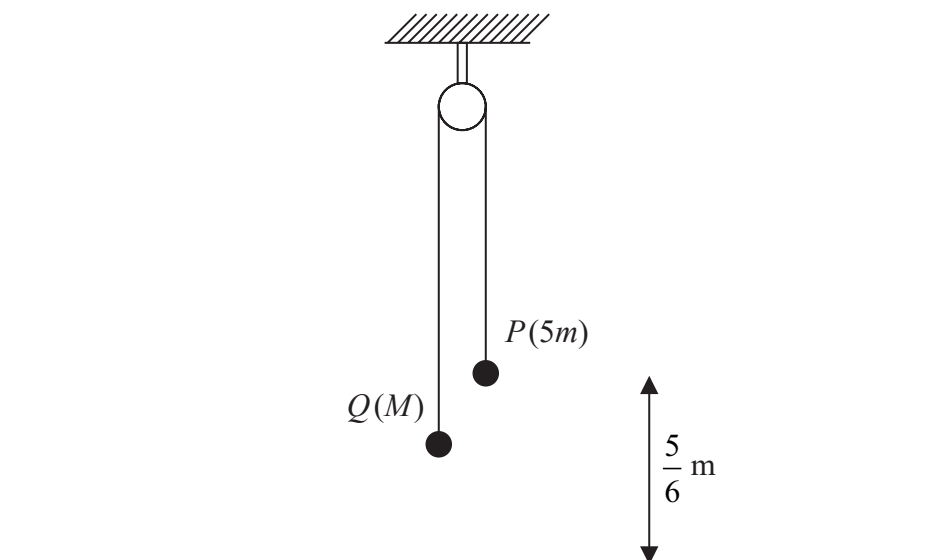


Figure 1

A particle P has mass $5m$.

A particle Q has mass M , where $M < 5m$.

The particles P and Q are connected by a light inextensible string.

The string passes over a small smooth light pulley that is fixed above horizontal ground.

Particle P is held at rest at a distance $\frac{5}{6} \text{ m}$ above the ground.

Particle Q hangs freely with the string taut, as shown in Figure 1.

At time $t = 0$, P is released from rest and begins to move downwards.

The particle P reaches the ground before Q reaches the pulley.

As P moves downwards, the tension in the string is $\frac{10mg}{3}$

In an initial model,

- air resistance is negligible
- P hits the ground at time $t = T$ seconds

Using the model,

(a) find M in terms of m (6)

(b) find the value of T (2)

In a refinement to the model,

- air resistance is included
- P hits the ground at time $t = T_1$ seconds

(c) State which is greater, T or T_1 . You must give a reason for your answer. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR PAPER IS 30 MARKS

